Assignment #3 - Team Contract and Individual Capstone Assessment
Muneer Al-Khasawneh

Our senior design project is developing a 3D Printing Lab Management Platform for the University of Cincinnati. The platform is based on 3D Print OS but customized for the needs of UC with features such as student submissions, manager approvals, and printer monitoring. As a Computer Science student, this project gives me the opportunity to apply software development and systems engineering concepts to create a real solution for the UC 3D print farm. My role will focus mainly on manager dashboards, maintenance utilities, and system monitoring. This matches closely with my background in programming, system design, and problem-solving courses. Overall, the project is a strong mix of technical depth and real-world application, providing a valuable service to UC students and faculty.

My coursework at the University of Cincinnati has prepared me well for this project. Data Structures and Intro to Computer Systems gave me the foundation for building efficient backend systems and understanding the interaction between software and hardware. Engineering & Design II provided experience in team-based projects, where we designed and built a fully functional robot programmed in Python. In addition, Intro Programming in Python and R introduced me to working with real datasets, which will be useful when adding analytics to the 3D printing pipeline. Together, these courses strengthened my ability to write efficient code, debug systems, and work collaboratively. I expect to apply these skills directly when building the dashboards and monitoring features of the system.

Outside of classwork, my job experience has also shaped how I will approach this project. As an Inventory Specialist at Parts of Moraine, I worked with large-scale cataloging, tracking, and management systems. This connects directly to the database and workflow aspects of our 3D printing platform. I also developed non-technical skills such as problem-solving, time management, and teamwork by supervising employees and expanding our sales to eBay. These experiences taught me the importance of scalability and ease of use in software systems. My professional background gives me the perspective to create features that are both useful and dependable for students and lab managers.

I am motivated to work on this project because it combines my interest in technology with a real application at my university. 3D printing is a fast-growing field, and designing a management system for UC gives me the chance to build something practical and lasting. I am especially interested in creating dashboards and monitoring tools that make the job easier for lab managers. This project is also an opportunity to showcase my skills in software engineering, system integration, and UI development. My motivation comes from knowing this project will not only help UC but also strengthen my portfolio for future opportunities. It is definitely motivating to potentially be creating something that could be beneficial to others.

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My approach is to break the system into modules: student submissions, approval workflows, printer monitoring, and analytics. By working step by step, I can make sure each module is reliable before integration. The expected result is a prototype that supports file submission, job approvals, and live printer monitoring, with room for future improvements. To evaluate my work, I will set weekly goals and test each feature thoroughly with my teammates before merging it. Success for me will be if lab managers can use the dashboards easily and the system runs smoothly without major issues. I will also measure my contributions based on team feedback and whether my work moves us closer to a complete, reliable platform.